# Digital Byzantine Studies - how Digital Humanities can help strengthen rare subjects

**Drach, Sviatoslav**
s.drach@uni-koeln.de
University of Cologne, Germany

**Neuefeind, Claes**
c.neuefeind@uni-koeln.de
University of Cologne, Germany

In recent years, the use of digital methods and tools has become an integral part of humanities research. However, this does not apply to all humanities disciplines to the same extent: smaller disciplines in particular run the risk of not keeping pace with the digital transformation. While some disciplines have developed a so-called "support field", as for example archaeoinformatics in the case of archaeology (cf. Schmidt / Hageneuer 2024), most small disciplines have to develop their own strategies. Using the example of Byzantine Studies, we want to discuss how small disciplines can be strengthened in the face of digital change. Based on the criteria of the 'Portal Kleine Fächer' ('Rare Subjects Portal') of the German Ministry of Culture, e.g. the small number of students and the low level of institutional representation, Byzantine Studies can be classified as a 'rare subject'. Like many other small disciplines, Byzantine Studies is challenged by structural shortcomings such as limited material and human resources and a scattered scholarly landscape. The main objective of this contribution is to argue that digital approaches are of key importance for the survival and development of the field and to show how techniques from Digital Humanities such as digital imaging, common encoding standards, and the use of dedicated digital infrastructures foster collaboration and multilinguality and thus strengthen the field as a whole (cf. Neuefeind et al. 2024).

In our contribution, we outline our efforts in the development of a digital infrastructure and the establishment of digital workflows and tools for Byzantine Studies with a specific focus on Byzantine sigillography. We report on our activities within the project " Digital Byzantine Studies (DiBS) – Creating a Sustainable Digital Infrastructure for Research-Based Teaching in Byzantine Studies " , carried out collaboratively by the Cologne Center for eHumanities ( CCeH ) and the Department of Byzantine Studies and Modern Greek Philology of the University of Cologne. The project is funded by the Volkswagen Foundation from 2022-2028 ( project no. 9B191 ).

The key objective of the project is to implement a set of strategic measures jointly developed by Byzantinists and digital humanists to strengthen the field of Byzantine Studies, with sigillography being one of four core areas of the field, alongside numismatics, epigraphy, and Greek manuscript studies. Over the first two years of the DiBS project, our focus has primarily been on integrating digital methods and tools into research practices in Byzantine Studies. These include:

- RTI (Reflectance Transformation Imaging), an imaging technology that enables images of small objects to be captured and displayed from different lighting perspectives. This raises the legibility of damaged seals (Schaeben / Catalano 2024), making it an important research tool for analysis.
- 3D imaging, a technology suitable not only for the presentation of seals, but also for 3D printing. Such printed seals can then be used for educational purposes while the original seals remain protected from potential damage.
- SigiDoc (Sopracasa et al. 2020), a semantic encoding standard based on Epidoc and optimized for Byzantine seals, developed collaboratively by the department for Byzantine Studies Cologne, the CCeH, and the CNRS (Paris).

Besides that, we focused on introducing innovative teaching and learning formats based on these digitally enhanced research methods within the existing study programmes in Cologne, taking a first step towards the institutionalization of Digital Byzantine Studies within an academic context. In this regard, the collaboration between the Department for Byzantine Studies and the CCeH has proven extremely fruitful, with mutual contributions to course offerings being well-received by students (Drach et al. 2024). As an example, a digital exhibition of around 35 seals was created as part of the seminar 'Approaching Byzantium Through Seals' which was held at the University of Cologne during winter 2023/2024. Not only did the students contribute to the development of the exhibition, but they also participated in its conception and shaped the resulting web presentation . The exhibition comprises digital images of seals (both RTI and 3D), which are accompanied by essays written by the students and reviewed by Byzantine experts with comments, interactive explanations and edited linked texts for each seal. It is

also possible to view the seals on a timeline and to view the seals in an embedded RTI and 3D viewer.

In our poster presentation, we describe some important steps taken towards a digital renewal of the discipline and discuss how technologies and methodologies from Digital Humanities can help overcome some of the challenges for Byzantine Studies that result from its status as a 'rare subject'.

# Bibliography

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